Revision Week

Emma James

Introduction

This session

We are now halfway through the module! Our timetabled session this week represents a chance to recap tricky concepts at your request, and then practice applying your new knowledge.

This worksheet has been designed in a way that reflects how you will need to use your knowledge in the exam, but with a more practical flavour. You might not get through all the exercises today (especially if we spend some time recapping content together), but you can use it as a revision tool and ask questions on the VLE.

The mock exam will be released on the VLE after the timetabled session (6pm, Monday), and will be available for you to do in your own time.

Approaching the worksheet

The worksheet will walk you through different analyses applied to a central (fictional) dataset. Where relevant, you can take your pick from three levels of R enthusiasm:

1. **(Wannabee) R Whizz!** Use your notes and analysis scripts from previous weeks to perform all the tasks from scratch, without looking at the code suggestions. Recommended for people who are independently motivated to learn and develop R skills.
2. **R Curious:** Click the "Show hint" tab and use the suggestions to set you on your way for each task. Recommended for people who are enjoying learning R but don't want to get held up at the cost of making it through the worksheet.
3. **R Overwhelmed:** Click the "Show code" tab, and copy and paste the code to generate output where necessary. Recommended for people who want to save learning R for another time.

You'll be able to explore these options via the tabs below each question, like this:

Don't help me!

Nothing to see here, you got this!

Show hint

Maybe I'll tell you the functions to use or remind you about key arguments.

Show code

For if you want to get straight to the output.

This means you can also move flexibly between these options: have a go at the parts you think you can, and don't be afraid to seek help when you need it. You can still ask questions in the practical session or on the VLE as usual, and navigate to earlier practicals via the "Weekly content" tab above.



Figure 1: You know some stuff! [Artwork by Allison Horst]

Fictional context

Today's dataset is simulated for the purpose of demonstrating the different analytic choices you might experience for any one project. Imagine that you are part of a university research team investigating student wellbeing. Your central question is: ***What helps students cope with stress, sleep better, and perform well academically?***

From your reading, you find that previous studies seem to disagree about which interventions work. You suspect that individual students might respond differently to interventions, and that the average effect may conceal important differences.

Set-up

- Open you **ARM_Practicals.RProj** file to get oriented in RStudio.
- Download the data from the VLE, and save it in your data folder as usual.
- Create a new R script for today's session, and save it in your scripts folder with a sensible name (e.g., ARM6_Revision)
- At the top of your script, load the following packages: *tidyverse, metafor, lme4, lmerTest, quantreg.* Note that you may need to install them first if you are working on a new device.

Don't help me!

Show hint

Use `install.packages("packagename")` to install a package, and `library(packagename)` to load it.

Show code

```
# If you need to install any packages... e.g.,
# install.packages("package")

# Load packages
library(tidyverse)
library(metafor)
library(lme4)
library(lmerTest)
library(quantreg)
```

Part A

Your colleague has already conducted a systematic review on different types of mindfulness interventions that have been trialled in student populations, and extracted the relevant effect sizes (**ARM6_metadat.csv**).

Q1: Which type of meta-analysis model would be appropriate for these data, and why? Load the dataset and use the `rma()` function from the *metafor* package to run this model.

Don't help me!

Show hint

You can use either sampling variance OR standard error, and need to decide whether to change the default method. Remember you can call `help(rma)` to inspect the function documentation.

Show code

```
# Load data
meta_dat <- read.csv("data/ARM6_metadat.csv")

# Run meta-analysis
meta_mod <- rma(yi = effect_size, vi = sampling_variance,
                data = meta_dat, method = "REML")
```

Q2: Create a forest plot based on your model. Use it determine:

a. How many of the original studies showed a significant benefit of the intervention?
b. Which studies were given most and least weight in the model?
c. Based on the meta-analysis of the existing literature, do interventions seems to work?

Don't help me!

Show hint

Use the `forest()` function!

Show code

```
# Forest plot
forest(meta_mod)
```

Q3: Create a funnel plot based on your model. What do you conclude from this?

Don't help me!

Show hint

Use the `funnel()` function!

Show code

```
# Funnel plot
funnel(meta_mod)
```

Part B

Your own university has already collected some data from a pilot intervention called RESET. 526 students signed up for the programme from across 12 different courses, and 378 completed it. Measures of stress and were recorded at each time point (self-ratings from 0-100), and an additional measure of sleep quality was collected at the end.

Load and inspect the data (**ARM6_trialdat.csv**). You should see that it is in *long format*: each participant has one row per assessment point (two rows in total).

Q4: Test whether there was a change in stress levels between the two time points (i.e., before and after the intervention). Choose a model that can handle the repeated measurements for each student, and retain participants with missing data in the analysis. You want to additionally account for the fact that students on different courses may have experienced different academic challenges during this period, and might experience difference changes in stress.

a. What type of model should you use, and how should the model be specified?
b. What was the average self-rated stress score before the intervention, and how did it change between the two time points?
c. Which course showed the least change in stress levels over time?

Don't help me!

Show hint

The `lmer()` function requires a formula for your fixed effects, plus (+) random effects specified as `(1|grouping)` for intercepts or `(1+predictor|grouping)` for slopes.

In the practical, we plotted random effects with the following code `dotplot.ranef.mer(ranef(model))`.

Show code

```
# Load data
trial_dat <- read.csv("data/ARM6_trialdat.csv")

# Run model
change_mod <- lmer(stress ~ time + (1+time|student_id) + (1+time|course),
                   data = trial_dat)

# Print output
summary(change_mod)

# Explore random effects
dotplot.ranef.mer(ranef(change_mod))
```

Q5: You want to see whether stress levels at the end of the programme were affecting students' sleep. Run a both an ordinary linear regression model **and** median regression model to test whether stress levels predict sleep quality at the second test point.

a. Which model is more appropriate for this dataset?
b. What do you conclude from the results?

Don't help me!

Show hint

Use `lm()` to fit an ordinary least squares regression, and `rq()` to fit a median regression model (specifying *tau* appropriately).

Show code

```
# Fit ordinary least squares regression model
ols_mod <- lm(sleep_quality ~ stress, data = trial_dat)

# Inspect output
summary(ols_mod)

# Check distribution of residuals
hist(resid(ols_mod))

# Fit median regression model
med_mod <- rq(sleep_quality ~ stress, tau = 0.5, data = trial_dat)

# Inspect output
summary(med_mod)
```

Q6: You suspect that moderate levels of stress do not impact sleep for most people, but that it might have a substantial impact on the poorest sleepers. Extend your quantile regression model to test this hypothesis, and assess whether there is sufficient evidence in its favour.

Don't help me!

Show hint

You can fit more than one quantile at once using `tau = c(x, x, x)`, replacing the x values with your chosen quantiles. You can inspect the results further using `plot(summary(model))` and the `anova()` functions.

Show code

```
# Fit across different quantiles (feel free to change values of tau)
qr_mod <- rq(sleep_quality ~ stress, tau = c(0.1, 0.5, 0.9), data = trial_dat)

# Inspect output
summary(qr_mod)

# Test statistically
anova(qr_mod)
```

Q7: You are also interested in understanding more about the people who did versus did not complete the intervention (i.e., those who were missing data at the second time point). You have a hypothesis that the students who dropped out of the intervention had more timetabled hours per week. Run a t-test to test this hypothesis, and then calculate bootstrapped 95% confidence intervals to provide more robust estimates of the t-statistic in the context of a mild skew.

a. What does the main t-test show?
b. Do the bootstrapped confidence intervals support the same conclusions?

Don't help me!

Note, the best option is to resample the two groups independently so that ensure that enough participants in each group are sampled for each analysis.

Some example code to extract relevant observations from the completed group would be `filter(trial_dat, completed == 1) %>% pull(timetabled_hours)`.

Show hint

The best option is to resample the two groups independently (to ensure that enough participants are sampled in each analysis). Some example code to extract relevant observations from the completed group would be `filter(trial_dat, completed == 1) %>% pull(timetabled_hours)`. You should save the output to a new object, and then edit to do the same for the non-completed group.

Once you have your two variables to sample, you will need to:

i. Create a variable to store the t-statistics from your resampled t-tests.
ii. Set up a loop that runs e.g., 1000 times
iii. Within the loop: (1) resample each of your groups; (2) run a new t-test; and (3) save the t-statistics to the relevant part of your list.
iv. Use the `quantile()` function with `c(0.025, 0.975)` to calculate 95% confidence intervals.

Show code

```r
# Run t-test
t.test(timetabled_hours ~ completed, data = trial_dat)

# Split dataset into two groups, and extract the relevant variable for ease of resampling
comp_group <- filter(trial_dat, completed == 1) %>%
  pull(timetabled_hours)

notcomp_group <- filter(trial_dat, completed == 0) %>%
  pull(timetabled_hours)

# Set up vector to store t-statistics for each resampled t-test
all_t <- NULL

# Bootstrap
for (i in 1:1000){

  # Resample each dataset
  resample_comp <- sample(comp_group, replace = TRUE)
  resample_notcomp <- sample(notcomp_group, replace = TRUE)

  # Run new t-test
  new_test <- t.test(resample_comp, resample_notcomp)

  # Save t-statistic to list
  all_t[i] <- new_test$statistic
}

# Calculate 95% confidence intervals
quantile(all_t, c(0.025, 0.975))
```

Wrapping up

Great work! Don't worry if you didn't get through it all at once, all that analysis switching was quite the mind workout. This practical session was shiny and new this year, and was developed with your previous practical feedback in mind, so **please do let me know your thoughts** via the feedback survey below.

Remember that you will be able to access the mock exam on the VLE from **6pm Monday 9th November.** Good luck! See you for structural equation modelling in Week 7 :)

 How's it going?

Let us know how today went via this quick survey! You will need to be logged in with your university account to respond, but your submission will be anonymous.
For more general questions about the practical, please post them on the VLE Discussion Board.